1.48 Proof by Induction: Sums of Series

Worked Solutions

1 a)

$$\frac{1}{3}k(k+1)(k+2) + (k+1)(k+2) = (k+1)(k+2)\left(\frac{1}{3}k+1\right)$$
$$= \frac{1}{3}(k+1)(k+2)(k+3).$$

1 b)

$$\frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k(k+2)}{(k+1)(k+2)} + \frac{1}{(k+1)(k+2)}$$
$$= \frac{k(k+2)+1}{(k+1)(k+2)} = \frac{k^2+2k+1}{(k+1)(k+2)} = \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2} \text{ as req.}$$

1 c)

$$(k+1)! - 1 + (k+1)(k+1)! = (k+1)![1 + (k+1)] - 1$$
$$= (k+1)!(k+2) - 1 = (k+2)! - 1 \text{ as req.}$$

2 a)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} 2^{r-1} = 2^{1-1} = 1, \qquad 2^1 - 1 = 1,$$

so true for $n = 1$.

Assume true for $n = k$:

$$\sum_{r=1}^{k} 2^{r-1} = 2^k - 1.$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} 2^{r-1} = \sum_{r=1}^{k} 2^{r-1} + 2^{k+1-1}$$
$$= 2^k - 1 + 2^k = 2 \times 2^k - 1 = 2^{k+1} - 1,$$

which is $2^n - 1$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

2 b)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} 4^r = 4^1 = 4, \qquad \frac{4}{3}(4^1 - 1) = \frac{4}{3} \times 3 = 4,$$

so true for $n = 1$.

Assume true for $n = k$:

$$\sum_{r=1}^{k} 4^r = \frac{4}{3}(4^k - 1).$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} 4^r = \sum_{r=1}^{k} 4^r + 4^{k+1}$$

$$= \frac{4}{3}(4^k - 1) + 4 \times 4^k = \frac{4}{3}(4^k - 1) + \frac{4}{3} \times 3 \times 4^k$$

$$= \frac{4}{3}(4^k - 1 + 3 \times 4^k) = \frac{4}{3}(4 \times 4^k - 1) = \frac{4}{3}(4^{k+1} - 1),$$

which is $\frac{4}{3}(4^n - 1)$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

3 a)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} r = 1 = 1, \qquad \frac{1}{2} \times 1(1 + 1) = \frac{1}{2} \times 2 = 1,$$

so true for $n = 1$.

Assume true for $n = k$: $\displaystyle\sum_{r=1}^{k} r = \frac{1}{2}k(k + 1)$.

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} r = \sum_{r=1}^{k} r + (k + 1) = \frac{1}{2}k(k + 1) + (k + 1)$$

$$= \frac{1}{2}(k + 1)[k + 2] = \frac{1}{2}(k + 1)(k + 2),$$

which is $\frac{1}{2}n(n + 1)$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

3 b)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} r^2 = 1^2 = 1, \qquad \frac{1}{6} \times 1(1 + 1)(2 + 1) = \frac{1}{6} \times 2 \times 3 = 1,$$

so true for $n = 1$.

Assume true for $n = k$: $\displaystyle\sum_{r=1}^{k} r^2 = \frac{1}{6}k(k + 1)(2k + 1)$.

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} r^2 = \sum_{r=1}^{k} r^2 + (k + 1)^2$$

$$= \frac{1}{6}k(k + 1)(2k + 1) + (k + 1)^2 = \frac{1}{6}(k + 1)[k(2k + 1) + 6(k + 1)]$$

$$= \frac{1}{6}(k + 1)[2k^2 + 7k + 6] = \frac{1}{6}(k + 1)(k + 2)(2k + 3)$$

$$= \frac{1}{6}(k+1)(k+1+1)(2(k+1)+1),$$

which is $\frac{1}{6}n(n+1)(2n+1)$ with $k+1$ substituted in.

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n \geq 1$.

3 c)

Prove true for $n=1$:

$$\sum_{r=1}^{1} r^3 = 1^3 = 1, \qquad \frac{1}{4} \times 1^2(1+1)^2 = \frac{1}{4} \times 4 = 1,$$

so true for $n=1$.

Assume true for $n=k$: $\displaystyle\sum_{r=1}^{k} r^3 = \frac{1}{4}k^2(k+1)^2$.

Prove true for $n=k+1$:

$$\sum_{r=1}^{k+1} r^3 = \sum_{r=1}^{k} r^3 + (k+1)^3 = \frac{1}{4}k^2(k+1)^2 + (k+1)^3$$

$$= \frac{1}{4}(k+1)^2[k^2+4(k+1)] = \frac{1}{4}(k+1)^2[k^2+4k+4]$$

$$= \frac{1}{6}(k+1)^2(k+2)^2,$$

which is $\frac{1}{4}n^2(n+1)^2$ with $k+1$ substituted in.

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n \geq 1$.

4)

Prove true for $n=1$:

$$\sum_{r=1}^{1} r(r-1) = 1 \times 0 = 0, \qquad \frac{1}{3} \times 1(1-1)(1+1) = 0,$$

so true for $n=1$.

Assume true for $n=k$: $\displaystyle\sum_{r=1}^{k} r(r-1) = \frac{1}{3}k(k-1)(k+1)$.

Prove true for $n=k+1$:

$$\sum_{r=1}^{k+1} r(r-1) = \sum_{r=1}^{k} r(r-1) + (k+1)(k+1-1)$$

$$= \frac{1}{3}k(k-1)(k+1) + k(k+1) = \frac{1}{3}k(k+1)[(k-1)+3]$$

$$= \frac{1}{3}(k+1)k(k+2),$$

which is $\frac{1}{3}n(n-1)(n+1)$ with $k+1$ substituted in.

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n \geq 1$.

5)

$$-3-4-3+\cdots+n(n-4)=\sum_{r=1}^{n} r(r-4).$$

Prove true for $n=1$:

$$\sum_{r=1}^{1} r(r-4)=1\times -3=-3, \qquad \frac{1}{6}\times 1\times 2\times (2-11)=\frac{1}{6}\times 2\times -9=-3,$$

so true for $n=1$.

Assume true for $n=k$: $\displaystyle\sum_{r=1}^{k} r(r-4)=\frac{1}{6}k(k+1)(2k-11)$.

Prove true for $n=k+1$:

$$\begin{aligned}\sum_{r=1}^{k+1} r(r-4)&=\sum_{r=1}^{k} r(r-4)+(k+1)(k+1-4)\\ &=\frac{1}{6}k(k+1)(2k-11)+(k+1)(k-3)\\ &=\frac{1}{6}(k+1)[k(2k-11)+6(k-3)]\\ &=\frac{1}{6}(k+1)(2k^2-5k-18)=\frac{1}{3}(k+1)(k+2)(2k-9)\\ &=\frac{1}{6}(k+1)(k+2)(2(k+1)-11),\end{aligned}$$

which is $\frac{1}{6}n(n+1)(2n-11)$ with $k+1$ substituted in.

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n\geq 1$.

6)

He wrote $k+1$ rather than the $(k+1)$th term. So

$$\sum_{r=1}^{k+1} r(r+3)=\sum_{r=1}^{k} r(r+3)+(k+1)(k+4).$$

7)

Prove true for $n=1$:

$$\sum_{r=1}^{1}\frac{1}{(r+1)(r+2)}=\frac{1}{2\times 3}=\frac{1}{6}, \qquad \frac{1}{2+4}=\frac{1}{6},$$

so true for $n=1$.

Assume true for $n=k$: $\displaystyle\sum_{r=1}^{k}\frac{1}{(r+1)(r+2)}=\frac{k}{2k+4}$.

Prove true for $n=k+1$:

$$\begin{aligned}\sum_{r=1}^{k+1}\frac{1}{(r+1)(r+2)}&=\sum_{r=1}^{k}\frac{1}{(r+1)(r+2)}+\frac{1}{(k+1+1)(k+1+2)}\\ &=\frac{k}{2k+4}+\frac{1}{(k+2)(k+3)}\end{aligned}$$

$$= \frac{k(k+3)}{2(k+2)(k+3)} + \frac{2}{2(k+2)(k+3)}$$
$$= \frac{k^2+3k+2}{2(k+2)(k+3)} = \frac{(k+2)(k+1)}{2(k+2)(k+3)}$$
$$= \frac{k+1}{2(k+3)} = \frac{k+1}{2k+6} = \frac{k+1}{2(k+1)+4},$$

which is $\frac{n}{2n+4}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

8)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} \frac{1}{(r+2)(r+4)} = \frac{1}{3 \times 5} = \frac{1}{15}, \quad \frac{1(7+25)}{24 \times 4 \times 5} = \frac{32}{480} = \frac{1}{15},$$

so true for $n = 1$.

Assume true for $n = k$: $\displaystyle\sum_{r=1}^{k} \frac{1}{(r+2)(r+4)} = \frac{k(7k+25)}{24(k+3)(k+4)}$.

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} \frac{1}{(r+2)(r+4)} = \sum_{r=1}^{k} \frac{1}{(r+2)(r+4)} + \frac{1}{(k+1+2)(k+1+4)}$$
$$= \frac{k(7k+25)}{24(k+3)(k+4)} + \frac{1}{(k+3)(k+5)}$$
$$= \frac{k(7k+25)(k+5)}{24(k+3)(k+4)(k+5)} + \frac{24(k+4)}{24(k+3)(k+4)(k+5)}$$
$$= \frac{k(7k^2+60k+125)+24(k+4)}{24(k+3)(k+4)(k+5)}$$
$$= \frac{7k^3+60k^2+149k+96}{24(k+3)(k+4)(k+5)}$$
$$= \frac{(k+3)(7k^2+39k+32)}{24(k+3)(k+4)(k+5)}$$
$$= \frac{7k^2+39k+32}{24(k+4)(k+5)} = \frac{(k+1)(7k+32)}{24(k+4)(k+5)}$$
$$= \frac{(k+1)(7(k+1)+25)}{24(k+4)(k+5)},$$

which is $\frac{n(7n+25)}{24(n+3)(n+4)}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

9 a)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} \frac{1}{3^r} = \frac{1}{3}, \qquad \frac{1}{2} - \frac{1}{2 \times 3^1} = \frac{1}{2} - \frac{1}{6} = \frac{1}{3},$$

so true for $n = 1$.

Assume true for $n = k$: $\displaystyle\sum_{r=1}^{k} \frac{1}{3^r} = \frac{1}{2} - \frac{1}{2 \times 3^k}$.

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} \frac{1}{3^r} = \sum_{r=1}^{k} \frac{1}{3^r} + \frac{1}{3^{k+1}}$$

$$= \frac{1}{2} - \frac{1}{2 \times 3^k} + \frac{1}{3^{k+1}} = \frac{1}{2} - \frac{1}{2 \times 3 \times 3^k} + \frac{1}{3 \times 3^k}$$

$$= \frac{1}{2} - \frac{1}{6 \times 3^k} = \frac{1}{2} - \frac{1}{2 \times 3^{k+1}},$$

which is $\frac{1}{2} - \frac{1}{2 \times 3^n}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

9 b)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} \frac{r}{2^r} = \frac{1}{2^1} = \frac{1}{2}, \qquad 2 - \frac{1+2}{2^1} = 2 - \frac{3}{2} = \frac{1}{2},$$

so true for $n = 1$.

Assume true for $n = k$: $\displaystyle\sum_{r=1}^{k} \frac{r}{2^r} = 2 - \frac{k+2}{2^k}$.

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} \frac{r}{2^r} = \sum_{r=1}^{k} \frac{r}{2^r} + \frac{k+1}{2^{k+1}}$$

$$= 2 - \frac{k+2}{2^k} + \frac{k+1}{2^{k+1}} = 2 - \frac{2(k+2)}{2 \times 2^k} + \frac{k+1}{2^{k+1}}$$

$$= 2 + \frac{-2(k+2) + k + 1}{2^{k+1}} = 2 + \frac{-2k - 4 + k + 1}{2^{k+1}}$$

$$= 2 - \frac{k+3}{2^{k+1}},$$

which is $2 - \frac{n+2}{2^n}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

9 c)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} (r+1)2^r = 2 \times 2^1 = 4, \qquad 1 \times 2^{1+1} = 4,$$

so true for $n = 1$.

Assume true for $n = k$: $\displaystyle\sum_{r=1}^{k} (r+1)2^r = k \times 2^{k+1}$.

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1}(r+1)2^r = \sum_{r=1}^{k}(r+1)2^r + (k+1+1)2^{k+1}$$

$$= k \times 2^{k+1} + (k+1+1)2^{k+1} = 2k \times 2^{k+1} + 2 \times 2^{k+1}$$

$$= 2 \times 2^{k+1}(k+1) = 2^1 \times 2^{k+1}(k+1) = (k+1)2^{k+2},$$

which is $n \times 2^{n+1}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

10 a)

Set $n = 1$ so that $2^2 = \frac{1}{6}(a + b + c)$. Set $n = 2$ so that $2^2 + 3^2 = \frac{1}{3}(4a + 2b + c)$. Set $n = 3$ so that $2^2 + 3^2 + 4^2 = \frac{1}{2}(9a + 3b + c)$.

Which gives:

$$24 = a + b + c \quad (1)$$

$$39 = 4a + 2b + c \quad (2)$$

$$58 = 9a + 3b + c \quad (3)$$

Solving simultaneously:

$$(2) - (1): \quad 15 = 3a + b$$

$$(3) - (1): \quad 34 = 8a + 2b$$

So $a = 2$, $b = 9$; $c = 24 - 2 - 9 = 13$. So that $a = 2$, $b = 9$ and $c = 13$ as req.

10 b)

Prove true for $n = 1$:

$$\sum_{r=1}^{1}(r+1)^2 = 2^2 = 4, \qquad \frac{1}{6} \times 1(2 + 9 + 13) = \frac{1}{6} \times 24 = 4,$$

so true for $n = 1$.

Assume true for $n = k$: $\displaystyle\sum_{r=1}^{k}(r+1)^2 = \frac{1}{6}k(2k^2 + 9k + 13)$.

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1}(r+1)^2 = \sum_{r=1}^{k}(r+1)^2 + (k+2)^2$$

$$= \frac{1}{6}k(2k^2 + 9k + 13) + (k+2)^2$$

$$= \frac{1}{6}[k(2k^2 + 9k + 13) + 6(k+2)^2]$$

$$= \frac{1}{6}[2k^3 + 15k^2 + 37k + 24] = \frac{1}{6}(k+1)(2k^2 + 13k + 24)$$

$$= \frac{1}{6}(k+1)(2(k+1)^2 - 4k - 2 + 13k + 24)$$

$$= \frac{1}{6}(k+1)(2(k+1)^2 + 9k + 22)$$

$$= \frac{1}{6}(k+1)(2(k+1)^2 + 9(k+1) + 13),$$

which is $\frac{1}{6}n(2n^2+9n+13)$ with $k+1$ substituted in.

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n \geq 1$.

11)

Prove true for $n=1$:

$$\sum_{r=1}^{1}(r+i)i = -1+i, \qquad -1+\frac{1}{2}\times 1(1+1)i = -1+i,$$

so true for $n=1$.

Assume true for $n=k$: $\displaystyle\sum_{r=1}^{k}(r+i)i = -k+\frac{1}{2}k(k+1)i.$

Prove true for $n=k+1$:

$$\begin{aligned}\sum_{r=1}^{k+1}(r+i)i &= \sum_{r=1}^{k}(r+i)i + (k+1+i)i \\ &= -k+\frac{1}{2}k(k+1)i + (k+1+i)i \\ &= -k+\frac{1}{2}k^2i+\frac{1}{2}ki+ki+i-1 \\ &= -(k+1)+\left(\frac{1}{2}k^2+\frac{3}{2}k+1\right)i \\ &= -(k+1)+\frac{1}{2}(k^2+3k+2)i \\ &= -(k+1)+\frac{1}{2}(k+1)(k+2)i,\end{aligned}$$

which is $-n+\frac{1}{2}n(n+1)i$ with $k+1$ substituted in.

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n \geq 1$.

12)

Prove true for $n=1$:

$$\sum_{r=1}^{1} r\times r! = 1\times 1! = 1, \qquad (1+1)!-1 = 2-1 = 1,$$

so true for $n=1$.

Assume true for $n=k$: $\displaystyle\sum_{r=1}^{k} r\times r! = (k+1)!-1.$

Prove true for $n=k+1$:

$$\begin{aligned}\sum_{r=1}^{k+1} r\times r! &= \sum_{r=1}^{k} r\times r! + (k+1)\times(k+1)! \\ &= (k+1)!-1+(k+1)\times(k+1)! \\ &= [(k+1)!+(k+1)\times(k+1)!]-1 \\ &= (k+1)![(k+1)+1]-1 = (k+1)!(k+2)-1 = (k+2)!-1,\end{aligned}$$

which is $(n+1)!-1$ with $k+1$ substituted in.

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n \geq 1$.

13)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} \frac{r}{(r+1)!} = \frac{1}{2!} = \frac{1}{2}, \qquad 1 - \frac{1}{(1+1)!} = 1 - \frac{1}{2} = \frac{1}{2},$$

so true for $n = 1$.

Assume true for $n = k$: $\sum_{r=1}^{k} \frac{r}{(r+1)!} = 1 - \frac{1}{(k+1)!}$.

Prove true for $n = k + 1$:

$$\begin{aligned}
\sum_{r=1}^{k+1} \frac{r}{(r+1)!} &= \sum_{r=1}^{k} \frac{r}{(r+1)!} + \frac{k+1}{(k+2)!} \\
&= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!} \\
&= 1 - \frac{k+2}{(k+2)(k+1)!} + \frac{k+1}{(k+2)!} \\
&= 1 - \frac{k+2}{(k+2)!} + \frac{k+1}{(k+2)!} = 1 - \frac{1}{(k+2)!},
\end{aligned}$$

which is $1 - \frac{1}{(n+1)!}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

14)

Prove true for $n = 1$:

$$\begin{aligned}
\sum_{r=1}^{1} \frac{1}{(r+a)(r+a+1)} &= \frac{1}{(1+a)(2+a)} \\
\frac{1}{(1+a)(1+1+a)} &= \frac{1}{(1+a)(2+a)},
\end{aligned}$$

so true for $n = 1$.

Assume true for $n = k$: $\sum_{r=1}^{k} \frac{1}{(r+a)(r+a+1)} = \frac{k}{(1+a)(k+1+a)}$.

Prove true for $n = k + 1$:

$$\begin{aligned}
\sum_{r=1}^{k} \frac{1}{(r+a)(r+a+1)} &= \sum_{r=1}^{k} \frac{1}{(r+a)(r+a+1)} + \frac{1}{(k+1+a)(k+1+a+1)} \\
&= \frac{k}{(1+a)(k+1+a)} + \frac{1}{(k+1+a)(k+2+a)} \\
&= \frac{k(k+2+a)}{(1+a)(k+1+a)(k+2+a)} + \frac{1+a}{(1+a)(k+1+a)(k+2+a)} \\
&= \frac{k^2 + 2k + ak + 1 + a}{(1+a)(k+1+a)(k+2+a)} \\
&= \frac{(k+1+a)(k+1)}{(1+a)(k+1+a)(k+2+a)} \\
&= \frac{k+1}{(1+a)(k+2+a)},
\end{aligned}$$

which is $\frac{n}{(1+a)(n+1+a)}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

15)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} a^r = a^1 = a, \qquad \frac{a(a^1 - 1)}{a - 1} = \frac{a(a-1)}{a-1} = a,$$

so true for $n = 1$.

Assume true for $n = k$: $\displaystyle\sum_{r=1}^{k} a^r = \frac{a(a^k - 1)}{a - 1}$.

Prove true for $n = k + 1$:

$$\begin{aligned}
\sum_{r=1}^{k+1} a^r &= \sum_{r=1}^{k+1} a^r + (k+1)^r \\
&= \frac{a(a^k - 1)}{a - 1} + a^{k+1} \\
&= \frac{a(a^k - 1)}{a - 1} + \frac{(a-1)a^{k+1}}{a-1} \\
&= \frac{a \times a^k - a + a \times a^{k+1} - a^{k+1}}{a - 1} \\
&= \frac{a^{k+1} - a + a \times a^{k+1} - a^{k+1}}{a - 1} \\
&= \frac{-a + a \times a^{k+1}}{a - 1} = \frac{a(a^{k+1} - 1)}{a - 1},
\end{aligned}$$

which is $\frac{a(a^n-1)}{a-1}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.